Workshop – Week 2 – Some selected solutions

1. Convert the following binary numbers to decimal. Show all working.

(a) 1011 **(b)** 101010 **(c)** 110011

Solution to (a):
1011 = 8 x 1 + 4 x 0 + 2 x 1 + 1 x 1 = 8 + 0 + 2 + 1 = 11

Solution to (b):
101010 = 32 x 1 + 16 x 0 + 8 x 1 + 4 x 0 + 2 x 1 + 1 x 0 = 32 + 8 + 2 = 42

2. Convert the following decimal numbers to binary. Show all working.

(a) 17 **(b)** 22 **(c)** 35

Solution to (a):
17 / 2 = 8 remainder **1**, 8 / 2 = 4 rem **0**, 4 / 2 = 2 rem **0**, 2 / 2 = 1 rem **0**, 1 / 2 = 0 rem **1** (stop)
write down the remainders backwards = 10001

Solution to (b):
22 / 2 = 11 rem **0**, 11 / 2 = 5 rem **1**, 5 / 2 = 2 rem **1**, 2 / 2 = 1 rem **0**, 1 / 2 = 0 rem **1** (stop)
write down the remainders backwards = 10110

3. Convert the following binary numbers to hexadecimal. Show all working. You can show that your conversion is correct by converting both the binary and the hexadecimal to decimal.

(a) 1010101 **(b)** 111111 **(c)** 10011101

Solution to (a):
Add zeros on the left and make groups of 4 bits = 0101 0101
Write down the hex digit for each group of 4 bits = 55

Check your solution by converting both to decimal:
1010101 = 64 x 1 + 32 x 0 + 16 x 1 + 8 x 0 + 4 x 1 + 2 x 0 + 1 x 1 = 64 + 16 + 4 + 1 = 85 (decimal)
55 = 16 x 5 + 1 x 5 = 80 + 5 = 85 (decimal)

Solution to (b):
Add zeros on the left and make groups of 4 bits = 0011 1111
Write down the hex digit for each group of 4 bits = 3F

Check your solution by converting both to decimal:
111111 = 32 x 1 + 16 x 1 + 8 x 1 + 4 x 1 + 2 x 1 + 1 x 1 = 32 + 16 + 8 + 4 + 2 + 1 = 63 (decimal)
3F = 16 x 3 + 1 x 15 = 63

4. Convert the following hexadecimal numbers to binary. Show all working. You can show that your conversion is correct by converting both the hexadecimal and the binary to decimal.

(a) BE **(b)** 2A **(c)** 1C

Solution to (a):
Write each hexadecimal digit as a group of 4 binary digits = 1011 1110

Check your solution by converting both to decimal:
BE = 16 x 11 + 1 x 14 = 176 + 15 = 190 (decimal)
1011 1110 = 128 + 64 x 0 + 32 + 16 + 8 + 4 + 2 + 1 x 0 = 190 (decimal)

Solution to (b):
Write each hexadecimal digit as a group of 4 binary digits = 0010 1010

Check your solution by converting both to decimal:
2A = 16 x 2 + 1 x 10 = 42 (decimal)
00101010 = 32 x 1 + 16 x 0 + 8 x 1 + 4 x 0 + 2 x 1 + 1 x 0 = 32 + 8 + 2 = 42 (decimal)

5. Perform each of the following subtractions using twos complement arithmetic and a word size of 8 bits. Show all working and check your answers.

(a) 7 - 3 **(b)** 15 - 17 **(c)** 28 - 20

Solution to (a):
7 in binary (with 8 bits) = 0000 0111
3 in binary (with 8 bits) = 0000 0011
complement of 3 = 1111 1100
-3 (= complement + 1) = 1111 1101

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 7 =  0000 0111
-3 =  1111 1101
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      0000 0100
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There is an extra 1 that gets carried out on the left but we only have 8 bits so it disappears.

Result is 4. Thus 7 + (-3) = 4.

6. The following bit patterns represent values stored in twos complement notation in a computer with a word length of 8 bits. Calculate the decimal value in each case. Show all working.

(a) 0000 1010 **(b)** 1100 1100 **(c)** 0111 1111 **(d)** 1111 0101

Solution to (a):
0000 1010 This has 0 in the sign bit so it is a positive number = 8 x 1 + 4 x 0 + 2 x 1 + 1 x 0 = 10

Solution to (b):
1100 1100 This has 1 in the sign bit so it is a negative number. Use the twos complement process to convert it to a positive number.

Complement the number to get 0011 0011
Add 1 to the complement to get 0011 0100

This new number is 32 x 1 + 16 x 1 + 8 x 0 + 4 x 1 + 2 x 0 + 1 x 0 = 32 + 16 + 4 = 52.

Thus the original number was -52 because we have shown that – (–52) = 52.

***8.** Assume that a computer stores integer numbers in twos complement notation. What are the largest and smallest numbers that can be stored if the computer has a word length of:

(a) four bits **(b)** six bits **(c)** eight bits

Solution to (a):
four bits (including a sign bit) means largest number = 0111 = 7